

Role-based access control (RBAC) means users only get access to the parts of the system that match their role. In your project, a student, organization officer, OSA administrator, and system administrator do not have the same permissions. This is used so people cannot open, edit, or manage information that is not part of their responsibility.

Secure login and authentication system means the system checks first if the user is really allowed to enter. Users must log in using their account credentials, and the system protects passwords through secure methods like hashing. For higher-level users like administrators, extra protection such as MFA may also be added. This helps stop unauthorized users from entering the system.

Encrypted data transmission (HTTPS) means the information sent between the user's device and the website is protected while it is traveling through the internet. This prevents attackers from easily reading or stealing sensitive data like usernames, passwords, or submitted records.

Centralized and secure database storage means the system keeps important records in one organized database instead of having them scattered in papers or separate files. Since the database is centralized, it is easier to manage, monitor, retrieve, and protect records. Security measures are also applied so the stored data stays safe from unauthorized access or loss.

Compliance with data privacy and institutional policies means the system is designed to follow rules and standards about how data should be handled. In your capstone, this includes protecting student information, limiting access to authorized users, collecting only needed data, and supporting policies related to digital governance and privacy. This makes the system not only functional, but also legally and institutionally responsible.

A very easy way to remember them is:

RBAC = who can access what

Authentication = proving who the user is

HTTPS = protecting data while sending it

Secure database = protecting data while storing it

Compliance = following rules and policies

For defense, you can say:

These features were included to ensure that the system is secure, organized, and policy-compliant. RBAC controls user permissions, authentication verifies identity, HTTPS protects transmitted data, centralized secure storage protects records, and compliance ensures that the system respects privacy and institutional regulations.

